

Role-Partitioned Swarm Intelligence for Resilient UAV Coordination in Contested Electromagnetic Environments: A Soccer Field Swarm Formation Framework

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Abstract

The deployment of autonomous Unmanned Aerial Vehicle (UAV) swarms in contested electromagnetic environments presents significant challenges in coordination, scalability, communication resilience, and safety assurance. Existing swarm-control architectures predominantly rely on homogeneous coordination strategies in which all agents execute identical interaction policies while simultaneously pursuing multiple mission objectives. Although effective under nominal operating conditions, such approaches frequently suffer from resource-allocation conflicts, communication dependence, and reduced robustness under electronic warfare (EW) interference, packet loss, and network fragmentation. Despite substantial advances in distributed control and swarm intelligence, the role of structured functional specialisation within UAV swarms remains insufficiently explored.

This paper introduces Role-Partitioned Swarm Intelligence (RPSI), a novel swarm-coordination paradigm that decomposes a homogeneous UAV swarm into specialised functional sub-swarms with distinct operational responsibilities. To realise this concept, we propose the Soccer Field Swarm Formation (SFSF) framework, which organises agents into Defender, Midfielder, and Forward classes responsible for asset protection, threat interception, communication bridging, and offensive engagement. By distributing mission objectives across multiple operational echelons, the proposed framework transforms a high-dimensional coordination problem into a set of coupled lower-dimensional sub-problems, improving scalability and operational resilience.

A formal mathematical framework is developed on the Special Euclidean Group $SE(3)$ to model role-conditioned swarm dynamics. Distributed target estimation is achieved through Covariance Intersection-based information fusion, ensuring estimator consistency under intermittent connectivity and unknown cross-correlations. Safety-critical operation is enforced through High-Order Control Barrier Function (HO-CBF) filtering integrated within a decentralised quadratic programming architecture. Furthermore, a communication-aware autonomy mechanism enables graceful degradation under electronic attack conditions while preserving mission continuity.

The proposed framework is evaluated using a high-fidelity Software-in-the-Loop (SITL) environment coupling PX4 autopilot firmware with realistic communication, sensing, and rigid-body dynamics models. Comparative experiments, scalability studies, communication-degradation analyses, and ablation investigations demonstrate that role-partitioned coordination significantly improves mission effectiveness, robustness, and asset-protection performance relative to conventional consensus-based architectures. The results indicate that structured functional decomposition constitutes a promising direction for resilient next-generation UAV swarm operations in communication-contested environments.

Index Terms

UAV swarms; swarm intelligence; multi-agent systems; role-partitioned coordination; distributed control; electronic warfare resilience; geometric control; control barrier functions; distributed estimation; cyber-physical systems; autonomous aerial systems; swarm robotics.